Compile results from simulations

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Results from linear confounding and no confounding

This file provides code for gathering simulation results and making plots.

```
n <- 400
n1 <- 20
E <- 10
R <- 5
nz <- 10
Qmat <- makeQ(20)
W <- eigen(Qmat)$values
reps <- 500

beta <- matrix(c(3,-3,2,3,5,4,-2,4,5,7,2,-3,3,4,6,1,-4,2,3,4,5,1,3,4,6,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,

namez <- c("MSM_H_10", "Naive", "Spatial+80", "Spatial+40", "USM_H_10")
sim_namez <- c("sim1", "sim2", "sim3", "sim4", "sim5")
```

- Setting 1, $\phi = 1, \beta_{XZ} = 1$
- Setting 2, $\phi = 1, \beta_{XZ} = 2$
- Setting 3, $\phi = 2, \beta_{XZ} = 1$
- Setting 4, $\phi = 2, \beta_{XZ} = 2$
- Setting 5, $\phi = 1, \beta_{XZ} = 0$ (no confounding)

The result outputs are not provided due to limits of file size on GitHub.

```
load("../simulation/sim1.RData")
load("../simulation/sim2.RData")
load("../simulation/sim3.RData")
load("../simulation/sim4.RData")
load("../simulation/sim5.RData")
```

Compile results from all settings

```
met <- list()
sim <- list(sim1, sim2, sim3, sim4, sim5)
for (i in 1:length(sim)){
  met[[i]] <- metrics_over_all2(sim[[i]], beta, 5, 5)
}
```

```r
m <- 5
MSE_nonzero <- combine_sim_res("MSE_ave1", m, namez)
MSE_zero <- combine_sim_res("MSE_ave0", m, namez)
Bias_nonzero <- combine_sim_res("bias_ave1", m, namez)
Bias_zero <- combine_sim_res("bias_ave0", m, namez)
Cov_nonzero <- combine_sim_res("cov_ave1", m, namez)
Cov_zero <- combine_sim_res("cov_ave0", m, namez)
SD_nonzero <- combine_sim_res("sd_ave1", m, namez)
SD_zero <- combine_sim_res("sd_ave0", m, namez)
LG_nonzero <- combine_sim_res("length_ave1", m, namez)
LG_zero <- combine_sim_res("length_ave0", m, namez)
```

Order the metrics

These tables are included in the appendix.

```r
switch <- function(metric){
  # metric2 <- rbind(metric[1,], metric[8,], metric[5:7,])
  metric2 <- rbind(metric[1,], metric[5,], metric[2:4,])
  rownames(metric2) <- c("MSM", "USM", "Naive", "Spatial+80", "Spatial+40")
  return(metric2)
}
MSE_nonzero2 <- switch(MSE_nonzero)
MSE_zero2 <- switch(MSE_zero)
Bias_nonzero2 <- switch(Bias_nonzero)
Bias_zero2 <- switch(Bias_zero)
Cov_nonzero2 <- switch(Cov_nonzero)
Cov_zero2 <- switch(Cov_zero)
SD_nonzero2 <- switch(SD_nonzero)
SD_zero2 <- switch(SD_zero)
LG_nonzero2 <- switch(LG_nonzero)
LG_zero2 <- switch(LG_zero)
```

```r
round(t(MSE_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.05 0.08  0.05       0.08       0.22
## setting 2 0.07 0.09  0.07       0.08       0.21
## setting 3 0.05 0.08  0.05       0.08       0.23
## setting 4 0.07 0.09  0.06       0.08       0.23
## setting 5 0.04 0.08  0.05       0.08       0.23
```

```r
round(t(MSE_zero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.02 0.02  0.07       0.10       0.29
## setting 2 0.05 0.03  0.09       0.10       0.27
## setting 3 0.02 0.02  0.06       0.11       0.29
## setting 4 0.04 0.03  0.08       0.11       0.29
## setting 5 0.01 0.01  0.06       0.11       0.29
```

```r
round(t(Bias_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.10 0.11  0.09       0.03       0.01
## setting 2 0.11 0.13  0.16       0.09       0.03
## setting 3 0.09 0.10  0.06       0.01       0.01
## setting 4 0.10 0.11  0.11       0.01       0.01
## setting 5 0.08 0.10  0.01       0.01       0.01
```

```r
round(t(Bias_zero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.03 0.04  0.12       0.05       0.03
## setting 2 0.04 0.06  0.19       0.11       0.06
## setting 3 0.01 0.02  0.08       0.01       0.02
## setting 4 0.01 0.04  0.14       0.02       0.02
## setting 5 0.01 0.01  0.01       0.01       0.02
```

```r
round(t(Cov_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.98 0.99  0.86       0.95       0.95
## setting 2 0.98 0.98  0.70       0.92       0.95
## setting 3 0.98 0.99  0.91       0.95       0.95
## setting 4 0.98 0.98  0.82       0.95       0.95
## setting 5 0.94 1.00  0.95       0.95       0.95
```

```r
round(t(Cov_zero2),2)
```

```
##           MSM USM Naive Spatial+80 Spatial+40
## setting 1   1   1  0.87       0.94       0.95
## setting 2   1   1  0.72       0.92       0.95
## setting 3   1   1  0.91       0.95       0.95
## setting 4   1   1  0.82       0.95       0.95
## setting 5   1   1  0.95       0.95       0.95
```

```r
round(t(SD_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.32 0.48  0.18       0.26       0.45
## setting 2 0.35 0.46  0.15       0.24       0.43
## setting 3 0.32 0.48  0.19       0.27       0.45
## setting 4 0.35 0.48  0.17       0.27       0.45
## setting 5 0.23 0.46  0.20       0.27       0.45
```

```r
round(t(SD_zero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.28 0.36  0.21       0.31       0.54
```

```
## setting 2 0.33 0.36  0.18       0.28       0.52
## setting 3 0.27 0.36  0.22       0.33       0.54
## setting 4 0.32 0.37  0.20       0.33       0.54
## setting 5 0.17 0.32  0.24       0.33       0.54
```

```r
round(t(LG_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 1.28 1.94  0.70       1.03       1.75
## setting 2 1.41 1.89  0.60       0.94       1.71
## setting 3 1.28 1.95  0.75       1.07       1.77
## setting 4 1.41 1.95  0.68       1.07       1.77
## setting 5 0.92 1.87  0.79       1.07       1.77
```

```r
round(t(LG_zero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 1.13 1.50  0.82       1.23       2.10
## setting 2 1.35 1.50  0.69       1.11       2.03
## setting 3 1.11 1.50  0.88       1.29       2.12
## setting 4 1.32 1.53  0.79       1.29       2.12
## setting 5 0.66 1.35  0.95       1.29       2.12
```

sim results combining zero and nonzero

```r
met <- list()
sim <- list(sim1, sim2, sim3, sim4, sim5)
for (i in 1:length(sim)){
  met[[i]] <- metrics_over_all3(sim[[i]], beta, 5)
}
```

```r
MSE <- combine_sim_res("MSE_ave1", m, namez)
Bias <- combine_sim_res("bias_ave1", m, namez)
Cov <- combine_sim_res("cov_ave1", m, namez)
SD <- combine_sim_res("sd_ave1", m, namez)
LG <- combine_sim_res("length_ave1", m, namez)
```

```r
MSE2 <- switch(MSE)
Bias2 <- switch(Bias)
Cov2 <- switch(Cov)
SD2 <- switch(SD)
LG2 <- switch(LG)
```

Figure for the results for linear and no confounding

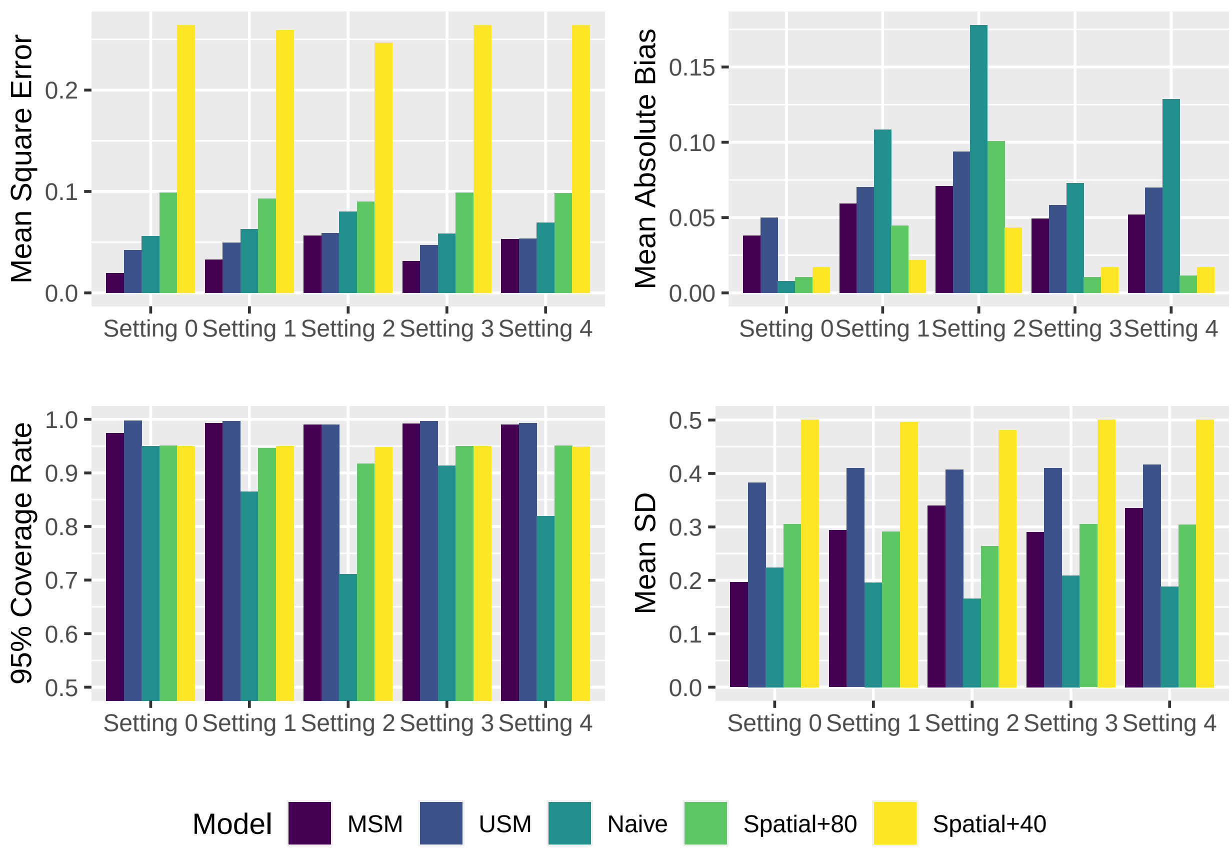
```r
# change no confounding to setting 0
# this plot doesn't include nonlinear confounding
plot_result9 <- function(metric, name){
```

```
  library(tidyverse)
  res <- data.frame(metric)
  res$model <- rownames(metric)
  res <- res %>% mutate(Model=factor(res$model, levels=c("MSM", "USM", "Naive", "Spatial+80", "Spatial+4
  res_v2 <- res %>% dplyr::rename(Setting1 = setting.1, Setting2 = setting.2, Setting3 = setting.3, Sett
  # , setting4 = setting.4
  res_v3 <- tidyr::pivot_longer(res_v2, cols = Setting1:Setting5, names_to = "Setting", values_to = name
  res_v3$new_setting <- revalue(res_v3$Setting, c("Setting5" = "Setting 0",
                                        "Setting1" = "Setting 1",
                                        "Setting2" = "Setting 2",
                                        "Setting3" = "Setting 3",
                                        "Setting4" = "Setting 4"))
  if (name == "95% Coverage Rate"){
    ggplot(data = res_v3, aes(x = new_setting, y = .data[[name]], fill = Model)) + geom_bar(stat = "iden
  }
  else {
    ggplot(data = res_v3, aes(x = new_setting, y = .data[[name]], fill = Model)) + geom_bar(stat = "iden
  }
}

metricz_name <- c("Mean Square Error", "Mean Absolute Bias", "95% Coverage Rate", "Mean SD", "mean LG")
metricz <- list(MSE2, Bias2, Cov2, SD2, LG2)
plotz <- list()
for (i in 1:4){
  plotz[[i]] <- plot_result9(metricz[[i]], metricz_name[i])
}
```

```
## Warning: package 'tidyverse' was built under R version 4.2.3

## Warning: package 'ggplot2' was built under R version 4.2.3

## Warning: package 'tibble' was built under R version 4.2.3

## Warning: package 'tidyr' was built under R version 4.2.3

## Warning: package 'readr' was built under R version 4.2.3

## Warning: package 'purrr' was built under R version 4.2.3

## Warning: package 'dplyr' was built under R version 4.2.3

## Warning: package 'stringr' was built under R version 4.2.3

## Warning: package 'forcats' was built under R version 4.2.3

## Warning: package 'lubridate' was built under R version 4.2.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.4
## v forcats   1.0.0     v stringr   1.5.1
```

```
## v ggplot2   3.4.4     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.0
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x purrr::accumulate()   masks foreach::accumulate()
## x dplyr::arrange()      masks plyr::arrange()
## x purrr::compact()      masks plyr::compact()
## x dplyr::count()        masks plyr::count()
## x dplyr::desc()         masks plyr::desc()
## x lubridate::dst()      masks LaplacesDemon::dst()
## x dplyr::failwith()     masks plyr::failwith()
## x dplyr::filter()       masks stats::filter()
## x dplyr::id()           masks plyr::id()
## x lubridate::interval() masks LaplacesDemon::interval()
## x dplyr::lag()          masks stats::lag()
## x dplyr::mutate()       masks plyr::mutate()
## x purrr::partial()      masks LaplacesDemon::partial()
## x dplyr::rename()       masks plyr::rename()
## x dplyr::select()       masks MASS::select()
## x dplyr::summarise()    masks plyr::summarise()
## x dplyr::summarize()    masks plyr::summarize()
## x purrr::when()         masks foreach::when()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```
allres <- ggpubr::ggarrange(plotz[[1]], plotz[[2]], plotz[[3]], plotz[[4]], common.legend = TRUE, legen
allres
```



```r
# ggsave("allres4.png")
```

Calculating Standard Error

```r
sim <- list(sim5, sim1, sim2, sim3, sim4)
se <- list()
for (i in 1:length(sim)){
  se[[i]] <- se_over_all2(sim[[i]], beta, 5, 5)
}
```

```r
m <- 5
MSE_se_nonzero <- combine_sim(se, "MSE_se_ave1", m, namez)
MSE_se_zero <- combine_sim(se, "MSE_se_ave0", m, namez)
Bias_se_nonzero <- combine_sim(se, "bias_se_ave1", m, namez)
Bias_se_zero <- combine_sim(se, "bias_se_ave0", m, namez)
Cov_se_nonzero <- combine_sim(se, "cov_se_ave1", m, namez)
Cov_se_zero <- combine_sim(se, "cov_se_ave0", m, namez)
SD_se_nonzero <- combine_sim(se, "sd_se_ave1", m, namez)
SD_se_zero <- combine_sim(se, "sd_se_ave0", m, namez)
LG_se_nonzero <- combine_sim(se, "length_se_ave1", m, namez)
LG_se_zero <- combine_sim(se, "length_se_ave0", m, namez)
```

Standard errors are included in the appendix

```r
print("MSE nonzero")
```

```
## [1] "MSE nonzero"
```

```r
MSE_se_nonzero
```

```
##              setting 1   setting 2   setting 3   setting 4   setting 5
## MSM_H_10   0.002087337 0.004254938 0.006257383 0.004046949 0.007237825
## Naive      0.003221166 0.003122201 0.003113914 0.003160087 0.003290423
## Spatial+80 0.005401735 0.005052791 0.004718543 0.005391191 0.005367142
## Spatial+40 0.014340015 0.014091860 0.013433818 0.014337383 0.014335558
## USM_H_10   0.006078602 0.006172707 0.006459053 0.006135427 0.006965561
```

```r
MSE_se_nonzero2 <- switch(MSE_se_nonzero)
round(t(MSE_se_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.00 0.01     0       0.01       0.01
## setting 2 0.00 0.01     0       0.01       0.01
## setting 3 0.01 0.01     0       0.00       0.01
## setting 4 0.00 0.01     0       0.01       0.01
## setting 5 0.01 0.01     0       0.01       0.01
```

```r
print("MSE zero")
```

```
## [1] "MSE zero"
```

```r
MSE_se_zero
```

```
##                setting 1   setting 2   setting 3   setting 4   setting 5
## MSM_H_10   0.0007976818 0.002389159 0.006092605 0.002379632 0.006772264
## Naive      0.0037672640 0.003868765 0.003910654 0.003862860 0.004071390
## Spatial+80 0.0068395114 0.006486446 0.006145766 0.006834705 0.006815941
## Spatial+40 0.0184556333 0.018076099 0.017153284 0.018453131 0.018450228
## USM_H_10   0.0023072188 0.003785492 0.004739920 0.003207168 0.004319367
```

```r
MSE_se_zero2 <- switch(MSE_se_zero)
round(t(MSE_se_zero2),2)
```

```
##            MSM USM Naive Spatial+80 Spatial+40
## setting 1 0.00   0     0       0.01       0.02
## setting 2 0.00   0     0       0.01       0.02
## setting 3 0.01   0     0       0.01       0.02
## setting 4 0.00   0     0       0.01       0.02
## setting 5 0.01   0     0       0.01       0.02
```

```r
print("Bias nonzero")
```

```
## [1] "Bias nonzero"
```

```r
Bias_se_nonzero
```

```
##               setting 1   setting 2   setting 3   setting 4   setting 5
## MSM_H_10   0.006524288 0.008011256 0.009722594 0.007876001 0.009868256
## Naive      0.009262927 0.008222596 0.007024658 0.008814079 0.008147483
## Spatial+80 0.012197848 0.011627475 0.010621866 0.012184644 0.012157594
## Spatial+40 0.020082974 0.019931376 0.019440113 0.020082448 0.020082133
## USM_H_10   0.011065616 0.011013291 0.010871566 0.011190280 0.011345923
```

```r
Bias_se_nonzero2 <- switch(Bias_se_nonzero)
round(t(Bias_se_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.01 0.01  0.01       0.01       0.02
## setting 2 0.01 0.01  0.01       0.01       0.02
## setting 3 0.01 0.01  0.01       0.01       0.02
## setting 4 0.01 0.01  0.01       0.01       0.02
## setting 5 0.01 0.01  0.01       0.01       0.02
```

```r
print("Bias zero")
```

```
## [1] "Bias zero"
```

```r
Bias_se_zero
```

```
##               setting 1   setting 2   setting 3   setting 4   setting 5
## MSM_H_10   0.003479072 0.005753095 0.008996769 0.005692155 0.008678891
## Naive      0.010830562 0.009571028 0.008194428 0.010281384 0.009459647
## Spatial+80 0.014773722 0.014077125 0.012735619 0.014759121 0.014728885
## Spatial+40 0.024111308 0.023828477 0.023095954 0.024108582 0.024106368
## USM_H_10   0.004637246 0.006068194 0.006852290 0.005835150 0.006686692
```

```r
Bias_se_zero2 <- switch(Bias_se_zero)
round(t(Bias_se_zero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.00 0.00  0.01       0.01       0.02
## setting 2 0.01 0.01  0.01       0.01       0.02
## setting 3 0.01 0.01  0.01       0.01       0.02
## setting 4 0.01 0.01  0.01       0.01       0.02
## setting 5 0.01 0.01  0.01       0.01       0.02
```

```r
print("SD nonzero")
```

```
## [1] "SD nonzero"
```

```r
SD_se_nonzero
```

```
##                setting 1    setting 2    setting 3    setting 4    setting 5
## MSM_H_10   0.0041841917 0.0055265938 0.0062177035 0.0057058768 0.0064578757
## Naive      0.0005761488 0.0006029886 0.0005510348 0.0006427798 0.0006783572
## Spatial+80 0.0007309170 0.0007074976 0.0006667943 0.0007318052 0.0007305223
## Spatial+40 0.0016818114 0.0016647401 0.0016275450 0.0016816078 0.0016819866
## USM_H_10   0.0074090270 0.0074932643 0.0075487951 0.0077573306 0.0081279577
```

```r
SD_se_nonzero2 <- switch(SD_se_nonzero)
round(t(SD_se_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.00 0.01     0          0          0
## setting 2 0.01 0.01     0          0          0
## setting 3 0.01 0.01     0          0          0
## setting 4 0.01 0.01     0          0          0
## setting 5 0.01 0.01     0          0          0
```

```r
print("SD zero")
```

```
## [1] "SD zero"
```

```r
SD_se_zero
```

```
##               setting 1    setting 2    setting 3    setting 4    setting 5
## MSM_H_10   0.0042527093 0.0064735446 0.0077806367 0.0065996832 0.0077854939
## Naive      0.0006956343 0.0007075432 0.0006328316 0.0007721038 0.0007960170
## Spatial+80 0.0008669638 0.0008435580 0.0007895109 0.0008668067 0.0008667749
## Spatial+40 0.0020193031 0.0020063930 0.0019577514 0.0020196412 0.0020201083
## USM_H_10   0.0063266536 0.0078357078 0.0079972869 0.0077747604 0.0086146778
```

```r
SD_se_zero2 <- switch(SD_se_zero)
round(t(SD_se_zero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.00 0.01     0          0          0
## setting 2 0.01 0.01     0          0          0
## setting 3 0.01 0.01     0          0          0
## setting 4 0.01 0.01     0          0          0
## setting 5 0.01 0.01     0          0          0
```

```r
print("Cov nonzero")
```

```
## [1] "Cov nonzero"
```

```r
Cov_se_nonzero
```

```
##              setting 1   setting 2   setting 3   setting 4   setting 5
## MSM_H_10   0.007266025 0.004440677 0.004824880 0.004489370 0.004741161
## Naive      0.009995660 0.014407788 0.016853298 0.012329124 0.015721058
## Spatial+80 0.009762641 0.009881459 0.011827430 0.009764761 0.009797398
## Spatial+40 0.009704949 0.009714094 0.009835446 0.009701886 0.009709677
## USM_H_10   0.002627166 0.002894866 0.004407662 0.002870061 0.004324994
```

```r
Cov_se_nonzero2 <- switch(Cov_se_nonzero)
round(t(Cov_se_nonzero2),2)
```

```
##            MSM USM Naive Spatial+80 Spatial+40
## setting 1 0.01   0  0.01       0.01       0.01
## setting 2 0.00   0  0.01       0.01       0.01
## setting 3 0.00   0  0.02       0.01       0.01
## setting 4 0.00   0  0.01       0.01       0.01
## setting 5 0.00   0  0.02       0.01       0.01
```

```r
print("Cov zero")
```

```
## [1] "Cov zero"
```

```r
Cov_se_zero
```

```
##               setting 1    setting 2   setting 3   setting 4   setting 5
## MSM_H_10   0.0006730237 0.0008582861 0.002004157 0.000240000 0.002033632
## Naive      0.0094858438 0.0138446861 0.015557930 0.012131205 0.015080199
## Spatial+80 0.0095984194 0.0102310433 0.011938791 0.009648344 0.009517799
## Spatial+40 0.0097825260 0.0097627678 0.009980433 0.009805168 0.009810911
## USM_H_10   0.0000000000 0.0000800000 0.000080000 0.000080000 0.000160000
```

```r
Cov_se_zero2 <- switch(Cov_se_zero)
round(t(Cov_se_zero2),2)
```

```
##           MSM USM Naive Spatial+80 Spatial+40
## setting 1   0   0  0.01       0.01       0.01
## setting 2   0   0  0.01       0.01       0.01
## setting 3   0   0  0.02       0.01       0.01
## setting 4   0   0  0.01       0.01       0.01
## setting 5   0   0  0.02       0.01       0.01
```

```r
print("LG nonzero")
```

```
## [1] "LG nonzero"
```

```r
LG_se_nonzero
```

```
##              setting 1   setting 2   setting 3   setting 4   setting 5
## MSM_H_10   0.017713376 0.022717546 0.025021062 0.023567626 0.026079639
## Naive      0.002290331 0.002389990 0.002175649 0.002546894 0.002674326
## Spatial+80 0.002913422 0.002818772 0.002657461 0.002916964 0.002905429
## Spatial+40 0.006648351 0.006578138 0.006448828 0.006648081 0.006648897
## USM_H_10   0.029806235 0.030061581 0.030042893 0.031083167 0.032243443
```

```r
LG_se_nonzero2 <- switch(LG_se_nonzero)
round(t(LG_se_nonzero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.02 0.03     0          0       0.01
## setting 2 0.02 0.03     0          0       0.01
## setting 3 0.03 0.03     0          0       0.01
## setting 4 0.02 0.03     0          0       0.01
## setting 5 0.03 0.03     0          0       0.01
```

```r
print("LG zero")
```

```
## [1] "LG zero"
```

```r
LG_se_zero
```

```
##              setting 1   setting 2   setting 3   setting 4   setting 5
## MSM_H_10   0.018155556 0.026679254 0.031222798 0.027238471 0.031283228
## Naive      0.002756961 0.002797613 0.002503827 0.003048281 0.003137719
## Spatial+80 0.003463877 0.003364853 0.003144137 0.003465958 0.003465973
## Spatial+40 0.007985093 0.007939692 0.007754867 0.007985905 0.007986919
## USM_H_10   0.026733316 0.032362275 0.032684491 0.032148820 0.035203546
```

```r
LG_se_zero2 <- switch(LG_se_zero)
round(t(LG_se_zero2),2)
```

```
##            MSM  USM Naive Spatial+80 Spatial+40
## setting 1 0.02 0.03     0          0       0.01
## setting 2 0.03 0.03     0          0       0.01
## setting 3 0.03 0.03     0          0       0.01
## setting 4 0.03 0.03     0          0       0.01
## setting 5 0.03 0.04     0          0       0.01
```

Use xtable to make tables

```
print(xtable(t(MSE_se_nonzero), type = "latex", caption = "MSE (nonzero)", label = "t:MSE_se1"), file =
print(xtable(t(Bias_se_nonzero), type = "latex", caption = "Bias (nonzero)", label = "t:bias_se1"), fil
print(xtable(t(SD_se_nonzero), type = "latex", caption = "SD (nonzero)", label = "t:sd_se1"), file = "ta
print(xtable(t(Cov_se_nonzero), type = "latex", caption = "Coverage (nonzero)", label = "t:cov_se1"), f
print(xtable(t(LG_se_nonzero), type = "latex", caption = "Length (nonzero)", label = "t:lg_se1"), file =
```

```
print(xtable(t(MSE_se_zero), type = "latex", caption = "MSE (zero)", label = "t:MSE_se0"), file = "tabl
print(xtable(t(Bias_se_zero), type = "latex", caption = "Bias (zero)", label = "t:bias_se0"), file = "ta
print(xtable(t(SD_se_zero), type = "latex", caption = "SD (zero)", label = "t:sd_se0"), file = "table15
print(xtable(t(Cov_se_zero), type = "latex", caption = "Coverage (zero)", label = "t:cov_se0"), file =
print(xtable(t(LG_se_zero), type = "latex", caption = "Length (zero)", label = "t:lg_se0"), file = "tabl
```

Results for just nonlinear simulation

```
n <- 400
n1 <- 20
E <- 10
R <- 5
nz <- 10
Qmat <- makeQ(20)
W <- eigen(Qmat)$values
reps <- 100

beta <- matrix(c(3,-3,2,3,5,4,-2,4,5,7,2,-3,3,4,6,1,-4,2,3,4,5,1,3,4,6,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,

namez <- c("MSM_H_10", "Naive")
sim_namez <- c("sim1", "sim2", "sim3", "sim4", "sim5", "sim6")
```

The result outputs are not provided due to limits of file size on GitHub.

```
load("../simulation/nonlinear_sim1.RData")
load("../simulation/nonlinear_sim2.RData")
load("../simulation/nonlinear_sim3.RData")
load("../simulation/nonlinear_sim4.RData")
load("../simulation/nonlinear_sim5.RData")
load("../simulation/nonlinear_sim6.RData")
```

Compile results from all four settings

```
met <- list()
sim <- list(nonlinear_sim1, nonlinear_sim2, nonlinear_sim3, nonlinear_sim4, nonlinear_sim5, nonlinear_si
for (i in 1:length(sim)){
  met[[i]] <- metrics_over_all2(sim[[i]], beta, 2, 5)
}
```

```
m <- 6
MSE_nonzero <- combine_sim_res("MSE_ave1", m, namez)
MSE_zero <- combine_sim_res("MSE_ave0", m, namez)
Bias_nonzero <- combine_sim_res("bias_ave1", m, namez)
Bias_zero <- combine_sim_res("bias_ave0", m, namez)
Cov_nonzero <- combine_sim_res("cov_ave1", m, namez)
Cov_zero <- combine_sim_res("cov_ave0", m, namez)
SD_nonzero <- combine_sim_res("sd_ave1", m, namez)
SD_zero <- combine_sim_res("sd_ave0", m, namez)
LG_nonzero <- combine_sim_res("length_ave1", m, namez)
LG_zero <- combine_sim_res("length_ave0", m, namez)
```

Order the metrics

These tables are included in the appendix.

```
switch <- function(metric){
  metric2 <- rbind(metric[1,], metric[2,])
  rownames(metric2) <- c("MSM", "Naive")
  return(metric2)
}
MSE_nonzero2 <- switch(MSE_nonzero)
MSE_zero2 <- switch(MSE_zero)
Bias_nonzero2 <- switch(Bias_nonzero)
Bias_zero2 <- switch(Bias_zero)
Cov_nonzero2 <- switch(Cov_nonzero)
Cov_zero2 <- switch(Cov_zero)
SD_nonzero2 <- switch(SD_nonzero)
SD_zero2 <- switch(SD_zero)
LG_nonzero2 <- switch(LG_nonzero)
LG_zero2 <- switch(LG_zero)
```

sim results combining zero and nonzero

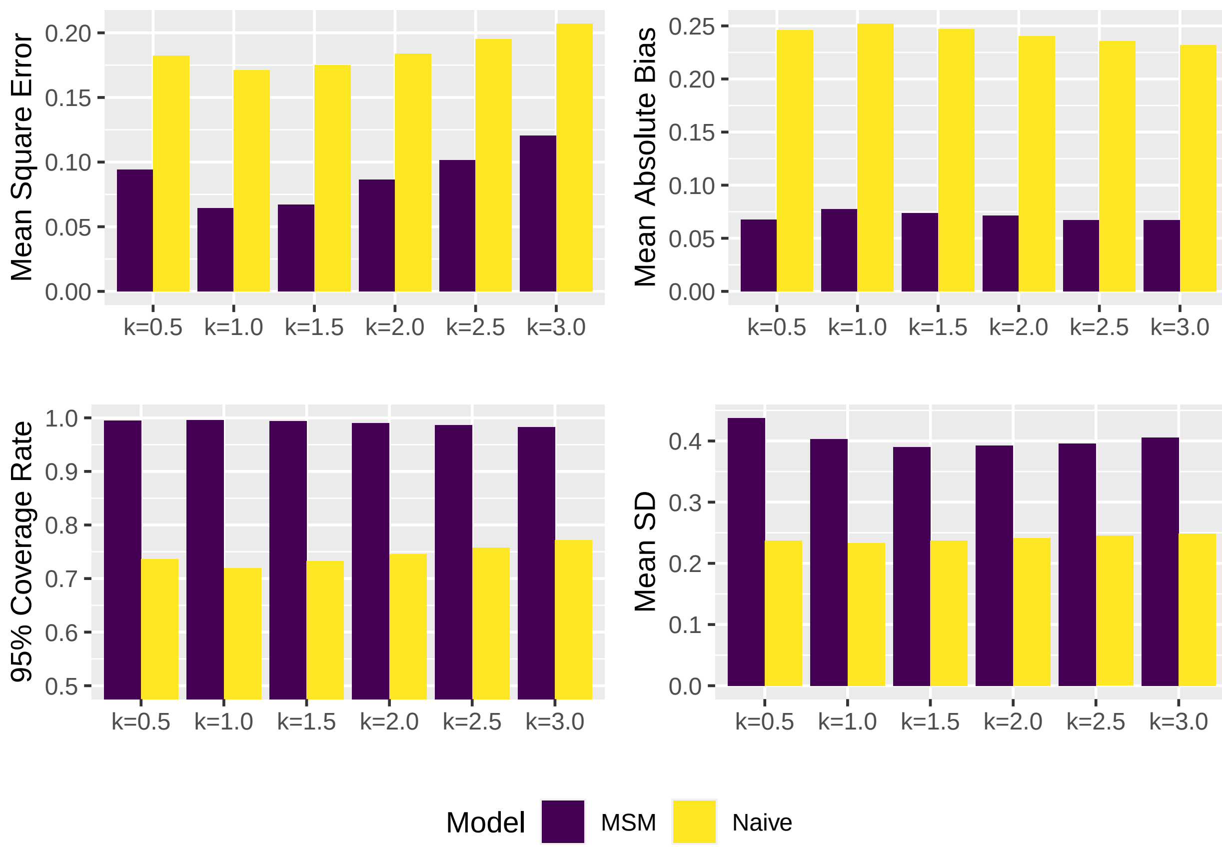
```
met <- list()
sim <- list(nonlinear_sim1, nonlinear_sim2, nonlinear_sim3, nonlinear_sim4, nonlinear_sim5, nonlinear_si
for (i in 1:length(sim)){
  met[[i]] <- metrics_over_all3(sim[[i]], beta, 2)
}
```

```
MSE <- combine_sim_res("MSE_ave1", m, namez)
Bias <- combine_sim_res("bias_ave1", m, namez)
Cov <- combine_sim_res("cov_ave1", m, namez)
SD <- combine_sim_res("sd_ave1", m, namez)
LG <- combine_sim_res("length_ave1", m, namez)
```

```
MSE2 <- switch(MSE)
Bias2 <- switch(Bias)
Cov2 <- switch(Cov)
SD2 <- switch(SD)
LG2 <- switch(LG)
```

Figure for the results for nonlinear confounding

```
plot_result10 <- function(metric, name){
  library(tidyverse)
  res <- data.frame(metric)
  res$model <- rownames(metric)
  res <- res %>% mutate(Model=factor(res$model, levels=c("MSM", "Naive")))
  res_v2 <- res %>% dplyr::rename(Setting1 = setting.1, Setting2 = setting.2, Setting3 = setting.3, Set
  # , setting4 = setting.4
  res_v3 <- tidyr::pivot_longer(res_v2, cols = Setting1:Setting6, names_to = "Setting", values_to = nam
  res_v3$new_setting <- revalue(res_v3$Setting, c("Setting1" = "k=0.5",
                                           "Setting2" = "k=1.0",
                                           "Setting3" = "k=1.5",
                                           "Setting4" = "k=2.0",
                                           "Setting5" = "k=2.5",
                                           "Setting6" = "k=3.0"))
  if (name == "95% Coverage Rate"){
    ggplot(data = res_v3, aes(x = new_setting, y = .data[[name]], fill = Model)) + geom_bar(stat = "ide
  }
  else {
    ggplot(data = res_v3, aes(x = new_setting, y = .data[[name]], fill = Model)) + geom_bar(stat = "ide
  }
}

metricz_name <- c("Mean Square Error", "Mean Absolute Bias", "95% Coverage Rate", "Mean SD", "mean LG")
metricz <- list(MSE2, Bias2, Cov2, SD2, LG2)
plotz <- list()
for (i in 1:4){
  plotz[[i]] <- plot_result10(metricz[[i]], metricz_name[i])
}
allres <- ggpubr::ggarrange(plotz[[1]], plotz[[2]], plotz[[3]], plotz[[4]], common.legend = TRUE, legen
allres
```




```r
# ggsave("allres5.png")
```

Sensitivity Tests L and K

The result outputs are not provided due to limits of file size on GitHub.

```r
sim <- list()
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L5_K2.RData")
sim_L5_K2 <- sim
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L5_K5.RData")
sim_L5_K5 <- sim
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L5_K10.RData")
sim_L5_K10 <- sim
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L10_K2.RData")
sim_L10_K2 <- sim
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L10_K5.RData")
sim_L10_K5 <- sim
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L10_K10.RData")
sim_L10_K10 <- sim
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L20_K2.RData")
sim_L20_K2 <- sim
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L20_K5.RData")
sim_L20_K5 <- sim
```

```r
load("C:/Users/shihn/Documents/research/simulation/setting19/sim_L20_K10.RData")
sim_L20_K10 <- sim
```

```r
res_LK <- array(NA, c(400,10,5,7,9,20))
res_LK[,,,,1,] <- sim_L5_K2[,,,,1,]
res_LK[,,,,2,] <- sim_L5_K5[,,,,1,]
res_LK[,,,,3,] <- sim_L5_K10[,,,,1,]
res_LK[,,,,4,] <- sim_L10_K2[,,,,1,]
res_LK[,,,,5,] <- sim_L10_K5[,,,,1,]
res_LK[,,,,6,] <- sim_L10_K10[,,,,1,]
res_LK[,,,,7,] <- sim_L20_K2[,,,,1,]
res_LK[,,,,8,] <- sim_L20_K5[,,,,1,]
res_LK[,,,,9,] <- sim_L20_K10[,,,,1,]
```

```r
met <- list()
sim <- list(res_LK, res_LK)
for (i in 1:length(sim)){
  met[[i]] <- metrics_over_all2(sim[[i]], beta, 9, 5)
}
```

```r
namez <- c("sim_L5_K2", "sim_L5_K5", "sim_L5_K10", "sim_L10_K2", "sim_L10_K5", "sim_L10_K10", "sim_L20_
m <- 2
MSE_nonzero <- combine_sim_res("MSE_ave1", m, namez)
MSE_zero <- combine_sim_res("MSE_ave0", m, namez)
Bias_nonzero <- combine_sim_res("bias_ave1", m, namez)
Bias_zero <- combine_sim_res("bias_ave0", m, namez)
Cov_nonzero <- combine_sim_res("cov_ave1", m, namez)
Cov_zero <- combine_sim_res("cov_ave0", m, namez)
SD_nonzero <- combine_sim_res("sd_ave1", m, namez)
SD_zero <- combine_sim_res("sd_ave0", m, namez)
LG_nonzero <- combine_sim_res("length_ave1", m, namez)
LG_zero <- combine_sim_res("length_ave0", m, namez)
```

```r
combined_LK <- cbind(MSE_nonzero[,1], MSE_zero[,1], Bias_nonzero[,1],Bias_zero[,1],Cov_nonzero[,1],Cov_
colnames(combined_LK) <- c("MSE_nonzero", "MSE_zero", "Bias_nonzero", "Bias_zero", "Cov_nonzero", "Cov_
round(combined_LK,4)
```

```
##             MSE_nonzero MSE_zero Bias_nonzero Bias_zero Cov_nonzero Cov_zero
## sim_L5_K2        0.0542   0.0242       0.1372    0.0521      0.8325    0.972
## sim_L5_K5        0.0496   0.0338       0.1157    0.0451      0.9725    1.000
## sim_L5_K10       0.0501   0.0555       0.1049    0.0475      0.9975    1.000
## sim_L10_K2       0.0562   0.0162       0.1399    0.0520      0.8800    0.970
## sim_L10_K5       0.0579   0.0304       0.0897    0.0268      0.9925    1.000
## sim_L10_K10      0.0527   0.0238       0.1080    0.0409      1.0000    1.000
## sim_L20_K2       0.0482   0.0141       0.1171    0.0468      0.9525    0.986
## sim_L20_K5       0.0496   0.0220       0.1089    0.0482      0.9975    1.000
## sim_L20_K10      0.0470   0.0209       0.1081    0.0386      1.0000    1.000
##             SD_nonzero SD_zero LG_nonzero LG_zero
## sim_L5_K2       0.2022  0.1844     0.7879  0.7214
## sim_L5_K5       0.3208  0.3263     1.2922  1.3199
## sim_L5_K10      0.3729  0.4048     1.5009  1.6332
## sim_L10_K2      0.2367  0.1887     0.9433  0.7675
```

```
## sim_L10_K5      0.3932  0.3566     1.5829  1.4511
## sim_L10_K10     0.4071  0.4147     1.6487  1.7034
## sim_L20_K2      0.2850  0.2242     1.1271  0.9039
## sim_L20_K5      0.3959  0.3762     1.5976  1.5366
## sim_L20_K10     0.4406  0.4464     1.7787  1.8305
```

```r
se <- list()
for (i in 1:length(sim)){
  se[[i]] <- se_over_all2(sim[[i]], beta, 9, 5)
}
```

```r
m <- 2
MSE_se_nonzero <- combine_sim(se, "MSE_se_ave1", m, namez)
MSE_se_zero <- combine_sim(se, "MSE_se_ave0", m, namez)
Bias_se_nonzero <- combine_sim(se, "bias_se_ave1", m, namez)
Bias_se_zero <- combine_sim(se, "bias_se_ave0", m, namez)
Cov_se_nonzero <- combine_sim(se, "cov_se_ave1", m, namez)
Cov_se_zero <- combine_sim(se, "cov_se_ave0", m, namez)
SD_se_nonzero <- combine_sim(se, "sd_se_ave1", m, namez)
SD_se_zero <- combine_sim(se, "sd_se_ave0", m, namez)
LG_se_nonzero <- combine_sim(se, "length_se_ave1", m, namez)
LG_se_zero <- combine_sim(se, "length_se_ave0", m, namez)
```

```r
print("MSE nonzero")
```

```
## [1] "MSE nonzero"
```

```r
MSE_se_nonzero
```

```
##               setting 1   setting 2
## sim_L5_K2   0.007738791 0.007738791
## sim_L5_K5   0.006840962 0.006840962
## sim_L5_K10  0.006769947 0.006769947
## sim_L10_K2  0.006571826 0.006571826
## sim_L10_K5  0.009508029 0.009508029
## sim_L10_K10 0.007988057 0.007988057
## sim_L20_K2  0.005815704 0.005815704
## sim_L20_K5  0.006324862 0.006324862
## sim_L20_K10 0.006069666 0.006069666
```

```r
print("MSE zero")
```

```
## [1] "MSE zero"
```

```r
MSE_se_zero
```

```
##               setting 1   setting 2
## sim_L5_K2   0.004243328 0.004243328
## sim_L5_K5   0.007843745 0.007843745
## sim_L5_K10  0.013259471 0.013259471
```

```
## sim_L10_K2  0.002725553 0.002725553
## sim_L10_K5  0.005187432 0.005187432
## sim_L10_K10 0.003757770 0.003757770
## sim_L20_K2  0.002523762 0.002523762
## sim_L20_K5  0.004162987 0.004162987
## sim_L20_K10 0.003092710 0.003092710
```

```r
print("Bias nonzero")
```

```
## [1] "Bias nonzero"
```

```r
Bias_se_nonzero
```

```
##              setting 1  setting 2
## sim_L5_K2   0.01644173 0.01644173
## sim_L5_K5   0.01661343 0.01661343
## sim_L5_K10  0.01773516 0.01773516
## sim_L10_K2  0.01542885 0.01542885
## sim_L10_K5  0.02015367 0.02015367
## sim_L10_K10 0.01815034 0.01815034
## sim_L20_K2  0.01636341 0.01636341
## sim_L20_K5  0.01725454 0.01725454
## sim_L20_K10 0.01712407 0.01712407
```

```r
print("Bias zero")
```

```
## [1] "Bias zero"
```

```r
Bias_se_zero
```

```
##              setting 1  setting 2
## sim_L5_K2   0.01394767 0.01394767
## sim_L5_K5   0.01691987 0.01691987
## sim_L5_K10  0.02211760 0.02211760
## sim_L10_K2  0.01011316 0.01011316
## sim_L10_K5  0.01653049 0.01653049
## sim_L10_K10 0.01408228 0.01408228
## sim_L20_K2  0.01018854 0.01018854
## sim_L20_K5  0.01324324 0.01324324
## sim_L20_K10 0.01296204 0.01296204
```

```r
print("Cov nonzero")
```

```
## [1] "Cov nonzero"
```

```r
Cov_se_nonzero
```

```
##               setting 1   setting 2
## sim_L5_K2   0.029506642 0.029506642
## sim_L5_K5   0.009381809 0.009381809
```

```
## sim_L5_K10  0.001118034 0.001118034
## sim_L10_K2  0.026518685 0.026518685
## sim_L10_K5  0.003354102 0.003354102
## sim_L10_K10 0.000000000 0.000000000
## sim_L20_K2  0.010403047 0.010403047
## sim_L20_K5  0.001118034 0.001118034
## sim_L20_K10 0.000000000 0.000000000
```

```r
print("Cov zero")
```

```
## [1] "Cov zero"
```

```r
Cov_se_zero
```

```
##               setting 1  setting 2
## sim_L5_K2   0.01140662 0.01140662
## sim_L5_K5   0.00000000 0.00000000
## sim_L5_K10  0.00000000 0.00000000
## sim_L10_K2  0.01164084 0.01164084
## sim_L10_K5  0.00000000 0.00000000
## sim_L10_K10 0.00000000 0.00000000
## sim_L20_K2  0.00570331 0.00570331
## sim_L20_K5  0.00000000 0.00000000
## sim_L20_K10 0.00000000 0.00000000
```

```r
print("SD nonzero")
```

```
## [1] "SD nonzero"
```

```r
SD_se_nonzero
```

```
##                setting 1   setting 2
## sim_L5_K2   0.009280896 0.009280896
## sim_L5_K5   0.012115304 0.012115304
## sim_L5_K10  0.011489989 0.011489989
## sim_L10_K2  0.009502590 0.009502590
## sim_L10_K5  0.015427964 0.015427964
## sim_L10_K10 0.012540100 0.012540100
## sim_L20_K2  0.010327351 0.010327351
## sim_L20_K5  0.012792210 0.012792210
## sim_L20_K10 0.012512420 0.012512420
```

```r
print("SD zero")
```

```
## [1] "SD zero"
```

```r
SD_se_zero
```

```
##              setting 1   setting 2
## sim_L5_K2   0.009773081 0.009773081
## sim_L5_K5   0.014008255 0.014008255
## sim_L5_K10  0.015609593 0.015609593
## sim_L10_K2  0.010479127 0.010479127
## sim_L10_K5  0.014467890 0.014467890
## sim_L10_K10 0.015480069 0.015480069
## sim_L20_K2  0.012519350 0.012519350
## sim_L20_K5  0.017121227 0.017121227
## sim_L20_K10 0.017171946 0.017171946
```

```
combined_LK_se <- cbind(MSE_se_nonzero[,1], MSE_se_zero[,1], Bias_se_nonzero[,1],Bias_se_zero[,1],Cov_se
colnames(combined_LK) <- c("MSE_nonzero", "MSE_zero", "Bias_nonzero", "Bias_zero", "Cov_nonzero", "Cov_
round(combined_LK_se,4)
```

```
##               [,1]   [,2]   [,3]   [,4]   [,5]   [,6]   [,7]   [,8]   [,9]
## sim_L5_K2   0.0077 0.0042 0.0164 0.0139 0.0295 0.0114 0.0093 0.0098 0.0371
## sim_L5_K5   0.0068 0.0078 0.0166 0.0169 0.0094 0.0000 0.0121 0.0140 0.0498
## sim_L5_K10  0.0068 0.0133 0.0177 0.0221 0.0011 0.0000 0.0115 0.0156 0.0457
## sim_L10_K2  0.0066 0.0027 0.0154 0.0101 0.0265 0.0116 0.0095 0.0105 0.0392
## sim_L10_K5  0.0095 0.0052 0.0202 0.0165 0.0034 0.0000 0.0154 0.0145 0.0621
## sim_L10_K10 0.0080 0.0038 0.0182 0.0141 0.0000 0.0000 0.0125 0.0155 0.0510
## sim_L20_K2  0.0058 0.0025 0.0164 0.0102 0.0104 0.0057 0.0103 0.0125 0.0419
## sim_L20_K5  0.0063 0.0042 0.0173 0.0132 0.0011 0.0000 0.0128 0.0171 0.0520
## sim_L20_K10 0.0061 0.0031 0.0171 0.0130 0.0000 0.0000 0.0125 0.0172 0.0506
##              [,10]
## sim_L5_K2   0.0384
## sim_L5_K5   0.0557
## sim_L5_K10  0.0608
## sim_L10_K2  0.0442
## sim_L10_K5  0.0586
## sim_L10_K10 0.0631
## sim_L20_K2  0.0517
## sim_L20_K5  0.0682
## sim_L20_K10 0.0689
```